

PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Pentester Name(Cybersecurity Professional)	Ngaakudzwe Salem Nyanduwa
Program/Batch	B083c-Networkwalks
Date	18 September 2026
Modules completed	W2-PM1 (Multiple Kali Tools) W2-PM5 (Zenmap Scanning)
Client/Target	1. Networkwalks (secured written permission already) 2. My own local LAN Network
Permission secured from client?	Yes
Phases covered	Phase 1: Reconnaissance & Footprinting Phase 2: Scanning & Network Discovery Phase 3-5: In Progress

1. Liability Disclaimer

I have performed these activities only on the systems & devices where I had secured written permission or the devices/systems that I own myself. All these materials are for education and research purpose only. Do not use anything from here to break the law. The instructor, the authors and Networkwalks are not responsible for what you do with this knowledge. Every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of your job and a permanent record. In most countries unauthorised access is a crime even when nothing is damaged.

2. Introduction

This Week 2 report documents two practical cybersecurity activities: web-domain footprinting using Kali Linux tools (W2-PM1) and network discovery using Zenmap (W2-PM5). The activities demonstrate how reconnaissance can be used to collect publicly available information and how network scanning can identify active devices within an authorized local environment. These exercises formed part of my ongoing cybersecurity internship at Networkwalks.

The footprinting activities were carried out in Kali Linux, while the Zenmap exercise was performed on a Windows PC with Zenmap installed. The report records the commands and observations from the practical exercises, together with supporting evidence and an explanation of the security relevance of each finding.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux & Windows	Operating systems used for reconnaissance activities
WHOIS	Find domain registration details (owner, dates, name servers).
whatweb	Fingerprint web technologies (server, CMS, plugins, IP).
nslookup	Resolve the domain name to its IP address using DNS.
curl -I	Read the HTTP response headers of the website.
wafw00f	Detect whether a Web Application Firewall protects the site.
dnsrecon	Enumerate all DNS records (NS, MX, SPF, TXT, SRV).
Zenmap (Nmap GUI)	Scan the local subnet to find live hosts, IPs and MAC addresses.
Windows CMD	Local IP and MAC address identification

4. Activities Performed

4.1 Footprinting & Reconnaissance

I conducted authorized reconnaissance of the networkwalks.com domain using six Kali Linux tools: WHOIS, WhatWeb, Nslookup, Curl, Wafw00f and DNSRecon. Each tool provided a different category of information that helped build an overall picture of the target's publicly exposed infrastructure.

I started with WHOIS to review publicly available domain-registration information and identify the domain's name servers. This helped provide an initial view of the domain and its hosting infrastructure.

I used WhatWeb to fingerprint the website and identify its technologies. The results identified WordPress 7.0.4 and WP Download Manager 3.3.58, together with other technical information exposed by the site.

Next, I used Nslookup to resolve the domain name and identify its associated IP address. The observed result was 192.232.216.135.

I used Curl with the -I option to inspect the HTTP response headers. The response provided additional technical information about the web application and showed the WordPress REST API endpoint /wp-json/.

I then used Wafw00f to check whether a Web Application Firewall was present. The result identified ModSecurity (SpiderLabs), providing an indication of the security technology protecting the web application.

Finally, I used DNSRecon to enumerate publicly available DNS records. The results included information about name servers, mail servers, SPF/TXT records, service records and DNS software.

4.2 Network Scanning with Zenmap

For the second activity, I used Zenmap to perform network discovery on my own local network. The practical required identification of the local IP address and subnet, discovery of live hosts, identification of IP and MAC addresses, and creation of a network topology.

I first used the Windows ipconfig command to determine my local IP address and LAN subnet. I entered the subnet into Zenmap and used a Ping Scan to identify active hosts on the authorized local network.

The example results provided in the practical identified three live hosts:

- 192.168.10.112
- 192.168.10.109
- 192.168.10.109

The example results also included 3 MAC addresses.

After the scan, I opened Zenmap's Topology section, enabled the legend, and saved the resulting network topology as a PDF in accordance with the practical requirements.

Note: The subnet, number of hosts, IP addresses and MAC addresses shown in the practical example should be replaced with my own verified network results and screenshots before final submission.

5. Risk Analysis / Impact

Based on the information collected during the footprinting and network scanning activities, I identified the following potential risks.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Web technology information exposed	WhatWeb identified WordPress and WP Download Manager	Attackers may use exposed technology/version information to identify software requiring further security review	● Medium
2	Server IP address identifiable	Nslookup resolved the domain to 192.232.216.135	Provides information about the network location of the web service	● Low
3	HTTP technical information exposed	Curl returned HTTP response headers and exposed /wp-json/	May assist technology fingerprinting and further enumeration	● Low
4	WAF technology identifiable	Wafw00f identified ModSecurity (SpiderLabs)	Reveals information about the web application's security architecture	● Low
5	DNS infrastructure information exposed	DNSRecon identified DNS, mail and service-related records	DNS information can help build a broader infrastructure profile	● Medium
6	Multiple live hosts visible on local network	Zenmap identified four live hosts in the example network	Unknown or unauthorized devices may potentially be present on a network	● Medium

Risk level key: ● Critical ● Medium ● Low

The risks above are observations from the footprinting and scanning exercises, not confirmed vulnerabilities.

The practical exercises primarily involved information gathering and host discovery. No exploitation or vulnerability validation was performed as part of these two modules.

Therefore, the presence of information such as a software version, IP address or DNS record does not by itself mean that the system is vulnerable. Further authorized security testing would be required to confirm any actual vulnerability.

6. Recommendations

Based on the observations from these activities, I recommend the following security improvements:

1. **Review publicly exposed technology information** Organizations should regularly review what information about their web technologies, CMS and plugins is publicly visible.
2. **Keep software updated** CMS platforms, plugins and other web technologies should be regularly updated and reviewed against current security advisories.
3. **Review HTTP headers** HTTP response headers should be reviewed to determine whether unnecessary technical information is being exposed.
4. **Review DNS records regularly** DNS records should be checked periodically to ensure that only required information and services are publicly exposed.
5. **Properly configure and monitor the WAF** Keep the WAF (ModSecurity) enabled and tuned, since it already blocks naive attacks.
6. **Perform regular internal network discovery** Organizations should periodically scan their own networks to identify active devices.
7. **Investigate unknown devices** Any unexpected device discovered during network scanning should be investigated and verified.
8. **Maintain network documentation** Network topology and device information should be documented and updated regularly.
9. **Perform security testing with authorization** Reconnaissance and scanning should only be performed against systems and networks where appropriate authorization has been provided.

7. Conclusion

During Week 2 of my Cybersecurity & Ethical Hacking internship, I completed practical exercises covering web footprinting, reconnaissance, and local network scanning.

During the footprinting activity, I used six Kali Linux tools to gather information about the target domain. The exercise improved my understanding of how WHOIS provides domain information, WhatWeb fingerprints web technologies, Nslookup resolves domains, Curl examines HTTP headers, Wafw00f identifies WAF technologies, and DNSRecon gathers DNS-related information.

During the network scanning activity, I used Zenmap to examine my local network configuration and identify active hosts. I also reviewed IP and MAC address information and generated a network topology.

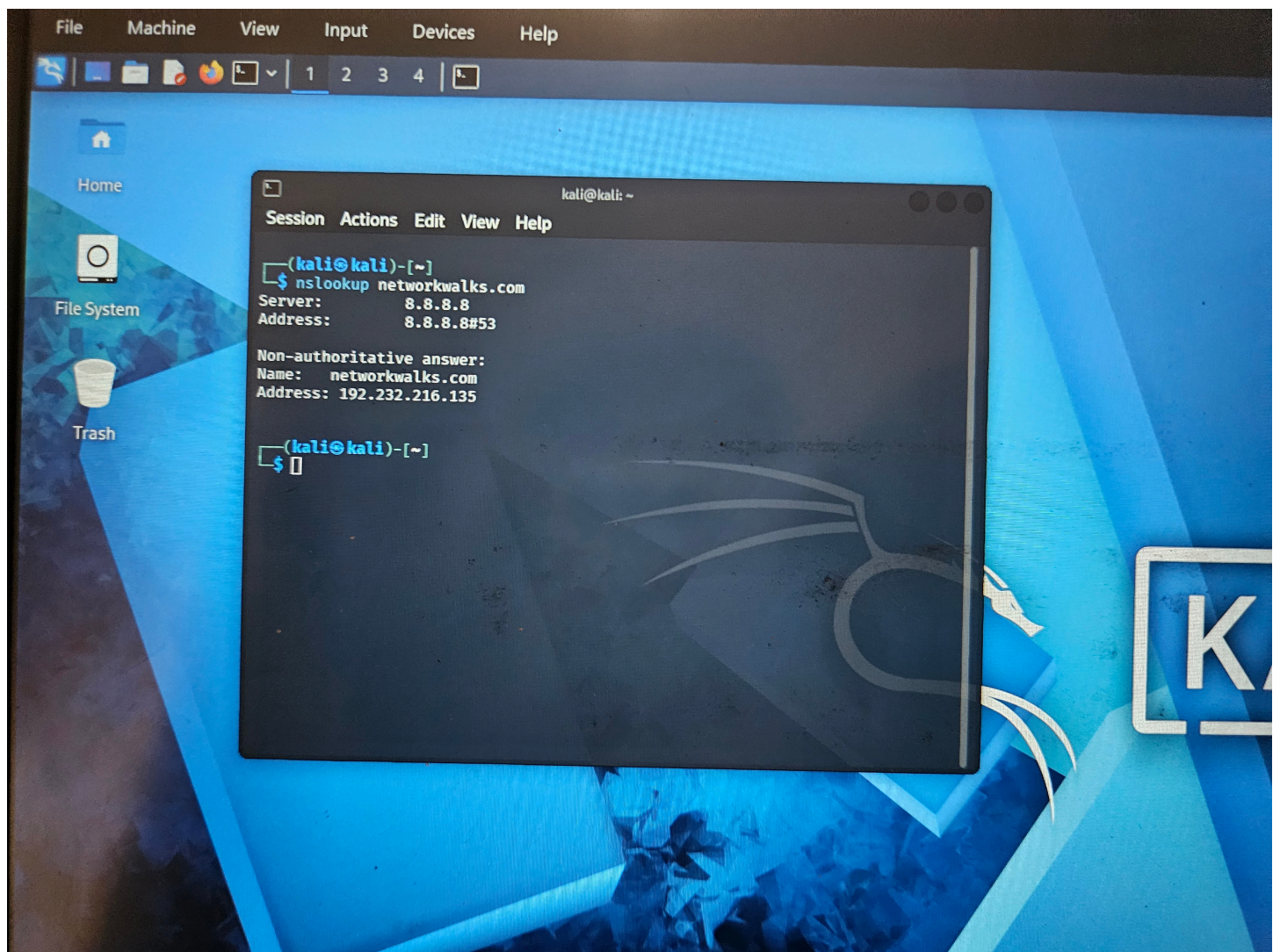
These exercises demonstrated to me that information gathering is an important part of cybersecurity. Before any exploitation is considered, a security professional can learn a considerable amount about an environment by analyzing publicly available information, DNS data, web responses, and network activity.

I also learned the importance of clear technical documentation. A useful cybersecurity report should state what was performed, record the findings and evidence, explain the security significance of each observation, and provide practical recommendations for reducing potential risk.

Finally, I reinforced the importance of performing reconnaissance and scanning only within an authorized scope. The activities documented in this report were completed as part of my assigned educational internship exercises.

8. Evidences Collected





Download the xDownload hist xInbox (18) - ng xINT - All Files x1 new messagexGitHub - Chanx+Ask Gemini

networkwalks.com/zenmap-network-scanning-practice-lab/

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Congratulations!

You have completed Zenmap Network Scanning (Practice Lab).

You have scored: **29%** (10/10)

A detailed copy of your quiz results has been sent to your email at: .

Please check your email & spam folder.

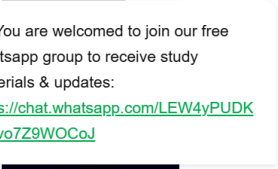
You can also **re-attempt** this quiz by reloading this page & new questions will appear from our question bank.

Your answers are shown below for self review & assessment:

CYBER IT DIPLOMA
NEW BATCH IS STARTING THIS MONTH
NETWORKWALKS.COM

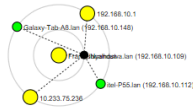
CCNA New Batch
Is starting
\$99

Hi, You are welcomed to join our free whatsapp group to receive study materials & updates:
<https://chat.whatsapp.com/LEW4yPUDK2J0vo7Z9WOCoj>



CORRECT ANSWERS:

```
Session Actions Edit View Help
kali@kali: ~
(kali@kali)-[~]
$ whatweb networkwalks.com
http://networkwalks.com [301 Moved Permanently] Apache, Cookies[__wpdm_client], Country[UNITED STATES][us], HTTPSer
32.216.135], RedirectLocation[https://networkwalks.com/], UncommonHeaders[permissions-policy,x-redirect-by,upgrade,
x-cache]
https://networkwalks.com [200 OK] Apache, Bootstrap[7.0.4], Cookies[__wpdm_client], Country[UNITED STATES][us], Ema
anager, HTML5, HTTPServer[Apache], HttpOnly[__wpdm_client], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordP
Open-Graph-Protocol[website], Script[4684NR-IPIB&pidnVar2=50511&pvtVar2=7&scvVar2=12,application/json,
xt/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endu
0.4]
https://networkwalks.com/ [200 OK] Apache, Bootstrap[7.0.4], Country[UNITED STATES][us], Email[info@networkwalks.co
er[Apache], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.0.4,WordPress Download Manager 3.3.58], O
B&pidnVar2=50511&pvtVar2=7&scvVar2=12,application/json,application/ld+json,module,speculationrules,text
ommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.0.
(kali@kali)-[~]
$
```



Zenmap

Scan Tools Profile Help

Target: 192.168.10.0/24 Profile: Quick scan Scan Cancel

Command: nmap -T4 -F 192.168.10.0/24

Hosts	Services	Nmap Output	Ports / Hosts	Topology	Host Details	Scans
OS	Host	nmap -T4 -F 192.168.10.0/24				
10.233.75.236		Starting Nmap 7.991 (https://nmap.org) at 2026-09-18 20:51 +0200				
192.168.10.1		Nmap scan report for 192.168.10.1				
Galaxy-S23.lan		Host is up (0.0063s latency).				
android-a3f15c		Not shown: 96 closed tcp ports (reset)				
FrancisNyanduwa.lan		PORT STATE SERVICE				
itel-P55.lan (192.168.10.112)		53/tcp open domain				
Galaxy-Tab-A8.lan		80/tcp open http				
HONOR-X7d-5		443/tcp open https				
		5000/tcp open upnp				
		MAC Address: 80:AF:CA:DA:6F:3C (Shenzhen Cudy Technology)				
		Nmap scan report for itel-P55.lan (192.168.10.112)				
		Host is up (0.012s latency).				
		All 100 scanned ports on itel-P55.lan (192.168.10.112) are in ignored states.				
		Not shown: 100 closed tcp ports (reset)				
		MAC Address: 8A:17:68:87:FA:A6 (Unknown)				
		Nmap scan report for FrancisNyanduwa.lan (192.168.10.109)				
		Host is up (0.0016s latency).				
		Not shown: 96 closed tcp ports (reset)				
		PORT STATE SERVICE				
		135/tcp open msrpc				
		139/tcp open netbios-ssn				
		445/tcp open microsoft-ds				
		7070/tcp open realserver				
		Nmap done: 256 IP addresses (3 hosts up) scanned in 8.53 seconds				

-End-

 Author

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Project Information

Program Name: Cybersecurity program at Networkwalks | **Week:** 02 | **Repository:** GitHub